

GARIMA AYACHIT

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PROFESSIONAL SUMMARY

Dynamic Bioinformatics scientist with over 8 years of comprehensive experience in NGS data analysis and AI/ML applications. Proficient in developing robust pipelines, conducting RNA-seq analysis, and applying advanced computational tools. Successfully engineered the ANNOTATOR-RNator pipeline and ML based genomic prediction pipeline miRDetect enhancing genomics and transcriptome analysis capabilities. Strong track record in mentoring and collaborating on complex projects. Adept at leveraging technical expertise to drive impactful research and innovation.

CORE SKILLS & EXPERTISE

- Bioinformatics & Genomics
 - NGS and Multi-omics (WGS, WES, RNA-seq, scRNA-seq, miRNA-seq, metagenomics) for eukaryotic, prokaryotic and fungal data
 - Comparative genomics, transcriptomics, and cancer biomarker discovery
 - Dual RNA-seq and host-pathogen interaction analysis
 - DNA barcoding and phylogenetics
- AI/ML, Programming & Data Science
 - Python (pipeline development, data processing, data analysis, biological resource mining and automation) pandas, numpy, scikit-learn
 - R (data visualization, statistical modeling, ggplot2, Tidyverse, DESeq2, normalization)
 - Machine learning approaches for spatio-temporal feature enhancement, classification and performance evaluation (SVM, Random Forest, kNN, cross-validation, ROC-AUC, feature importance and filtering and SHAP interpretability)
- High-Performance & Cloud Computing
 - HPC environments (SLURM, job scheduling, Singularity/apptainer containers), containerization
 - Compute Canada and Linux server administration
 - Cloud computing (AWS) for scalable bioinformatics workflows
- Tools & Pipelines
 - BWA, fastpGATK, STAR, HISAT2, DESeq2, Subread/FeatureCounts, Bedtools
 - Git/GitHub for version control and collaboration
 - Custom pipeline development (e.g., **ANNOTATOR-RNator**, **miRDetect**), command-line shell scripts
- Project Leadership & Collaboration
 - **Experience leading team** having wet-lab background and bioinformatician
 - Mentoring and training graduate students and researchers
 - Cross-disciplinary collaborations with PhD scientists and clinicians
 - Research proposal drafting (national/state-level grants)
 - Strong written and oral communication skills

PROFESSIONAL EXPERIENCE

Career Break (Maternity Leave) | 2024-2025

Postdoctoral Fellow | INRS, Laval, QC | Feb 2022 – Oct 2023

- **Engineered** ANNOTATOR-RNator pipeline for pan-species transcriptome analysis integrating BLAST, BWA, Prokka, NetworkX, Get_homologous, and FeatureCount
- **Executed** dual RNA-seq analysis of Neisseria-infected cell lines using DESeq2 to uncover host-pathogen interactions.
- **Leveraged** HPC (Compute Canada, SLURM) and Singularity for reproducible workflows and large-scale analysis.
- **Advised** in analysis of bulk Metagenome data using various tools
- **Guided** a doctoral student in scRNAseq analysis using Seurat
- **Implemented** shell scripting and version control with GitHub for reproducible workflows.

Research Bioinformatician | BaseSolve Informatics Pvt. Ltd | Oct 2020 – Feb 2021

- **Scrutinized** client NGS data (Eukaryotic WGS, Fungal genomics, amplicon sequence, RNAseq, exome)
- **Composed** detailed bioinformatics reports tailored to client research goals
- **Deployed** pipelines on SLURM clusters, Singularity containers, Linux environments, and AWS Cloud.

Research Associate | Gujarat Biotechnology Research Centre, India | July 2016 – July 2019

- **Led** team of junior research fellows for executing project on plant transcriptome analysis for upscaling of metabolites
- **Investigated** TCGA datasets to discover biomarkers for oral squamous cell carcinoma using differential expression analysis. Analyzed miRNA-seq data for different projects
- **Coordinated** and executed collaborative projects with IIT, India and trained students in Linux essentials
- **Drafted** research proposals for grant applications

Senior Research fellow | Gujarat State Biotechnology Mission, India | June 2014 – June 2016

- **Performed** whole-genome sequencing analysis of Asiatic lion (Schedule 1 species) to identify genetic variations.
- **Evaluated** WES and targeted cancer panels for rare disease research.
- **Characterized** saline desert microbiomes via shotgun metagenomic sequencing.
- **Generated** DNA barcodes for marine species biodiversity banking; deposited sequences in BOLD database.
- **Mentored** graduate students during Master's dissertation projects.

Junior Research fellow | Gujarat State Biotechnology Mission, India | Aug 2012- June 2014

- **Analyzed** whole-genome sequencing data of salt-tolerant desert bacteria.
- **Maintained** & customized PHP-MySQL grant management system for state research funding.

EDUCATIONAL CREDENTIALS

2022	PhD Bioinformatics , Gujarat University, India
2015	M. Phil. Biotechnology , Hemchandracharya North Gujarat University, Patan, Gujarat, India
2011	Master of Science , Bioinformatics, G.H Patel Department of Computer Science and Technology, (GDCST), Sardar Patel University, India
2009	Bachelor of Science , Bioinformatics, N.V Patel College of Pure and Applied Sciences, Sardar Patel University, India

SELECTED COMPUTATIONAL & ML PROJECTS

- **miRDetect**: Developed a Random Forest-based framework for automated detection of novel miRNA precursors; implemented feature selection, cross-validation, and ROC-based evaluation
- **ANNOTATOR-RNAtor**: Engineered a scalable transcriptome annotation pipeline integrating sequence alignment, homology search, and network-based analysis

PEER-REVIEWED PUBLICATIONS

- Giraud-Gatineau, A., **Ayachit, G.**, Nieves, C., Dagbo, K.C., Bourhy, K., Pulido, F., Huete, S.G., Benaroudj, N., Picardeau, M. and Veyrier, F.J., 2024. Inter-species transcriptomic analysis reveals a constitutive adaptation against oxidative stress for the highly virulent *Leptospira* species. **Molecular Biology and Evolution**, 41(4), p.msae066.
- **Ayachit, G.**, Pandya, H. and Das, J., 2020. miRDetect: A combinatorial approach for automated detection of novel miRNA precursors from plant EST data using homology and Random Forest classification. **Genomics**.
- **Ayachit, G.**, Shaikh, Inayatullah., Pandya, H. and Das, J., 2020. Salient Features, Data and Algorithms for microRNA Screening from Plants: A Review on the Gains and Pitfalls of Machine Learning Techniques. **Current Bioinformatics**.
- Kumar, S., **Ayachit, G.** and Sahoo, L., 2020. Screening of mungbean for drought tolerance and transcriptome profiling between drought-tolerant and susceptible genotype in response to drought stress. **Plant Physiology and**

Biochemistry, 157, pp.229-238.

- Lata, K.S., Vaghasia, V., Bhairappanavar, S.B., Kumar, S., **Ayachit, G.**, Patel, S. and Das, J., 2020. Whole genome sequencing and de novo assembly of three virulent Indian isolates of *Leptospira*. **Infection, Genetics and Evolution**, 85, p.104579.
- Mohanty, P., **Ayachit, G.**, Mohanty, J.N., Pandya, H., Mankad, A.U. and Das, J., 2020. Documentation of conserved and novel miRNAs participated in plant secondary metabolic pathways of sanctified *Aegle marmelos*. **Gene Reports**, 21, p.100943.
- Mohanty, P., **Ayachit, G.**, Sharma, P., Shaikh, I., Mohanty, J.N., Mankad, A.U., Pandya, H. and Das, J., 2020. De novo sequencing and transcriptome analysis of Indian Bael (*Aegle marmelos* L.). **Gene Reports**, 19, p.100671.
- **Ayachit, G.**, Shaikh, I., Sharma, P., Jani, B., Shukla, L., Sharma, P., Bhairappanavar, S.B., Joshi, C. and Das, J., 2019. De novo transcriptome of *Gymnema sylvestre* identified putative lncRNA and genes regulating terpenoid biosynthesis pathway. **Scientific Reports**, 9(1), pp.1-13.
- Shaikh, I., Ansari, A., **Ayachit, G.**, Gandhi, M., Sharma, P., Bhairappanavar, S., Joshi, C.G. and Das, J., 2019. Differential gene expression analysis of HNSCC tumors deciphered tobacco dependent and independent molecular signatures. **Oncotarget**, 10(58), pp.6168-6183.
- Trivedi, M.H., Bhatt, K.A., Pandit, A.S., Shukla, N.C., **Ayachit, G.N.**, Joshi, M.N., Tyagi, S.N., Bagatharia, S.B. and Bhandari, S., 2019. Evaluation and optimization of DNA extraction methods for benthic foraminifera from the Gulf of Kachchh, Gujarat, India. **Indian Journal of Experimental Biology**, 57(4).
- Kumar, D., Kumar, S., **Ayachit, G.**, Bhairappanavar, S.B., Ansari, A., Sharma, P., Soni, S. and Das, J., 2017. Cross-kingdom regulation of putative miRNAs derived from happy tree in cancer pathway: a systems biology approach. **International journal of molecular sciences**, 18(6), p.1191.
- Pandit, A.S., Joshi, M.N., Bhargava, P., Shaikh, I., **Ayachit, G.N.**, Raj, S.R., Saxena, A.K. and Bagatharia, S.B., 2015. A snapshot of microbial communities from the Kutch: one of the largest salt deserts in the World. **Extremophiles**, 19(5), pp.973-987.
- Mehta, A., Sinha, V.K. and **Ayachit, G.**, 2012. Land-use/land-cover study using remote sensing and GIS in an arid environment. **Bulletin of Environmental and Scientific Research**, 1(3-4), pp.4-8.

GITHUB PROJECT LINKS

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- <https://github.com/BactSymEvol/Annotator-RNAtor>